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# 3D Gaussian Splatting Reconstruction of UCLA Gateway Plaza for Modern Robotics Simulation

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## Abstract

As autonomous vehicle research increasingly relies on high-fidelity simulation, the need for realistic, interactive digital twins is steadily increasing. While neural implicit models have significantly advanced 3D scene reconstruction, their computational overhead often fails for real-time requiring sparse rendering for dynamic physics simulations. Explicit continuous scene representations, such as 3D Gaussian Splatting (3DGS), overcome these limitations by enabling photorealistic, real-time synthesis while retaining the ability for continuous spatial manipulation. This project presents an end-to-end 3DGS reconstruction of the UCLA Gateway Plaza, a highly trafficked pedestrian and vehicular intersection actively studied by the UCLA Mobility Lab for transportation research. By integrating this explicit representation into NVIDIA Isaac Sim alongside an aligned structural collision mesh, we establish a robust simulation environment. This hybrid approach provides autonomous vehicle algorithms with a rigorous physical collision map for kinematic control, as well as the rich perceptual, visual, and semantic features necessary to train and evaluate vision-based autonomous systems within a precise replica of UCLA Gateway Plaza.

## 1 Introduction

In this project, we create a high-fidelity digital twin of UCLA Gateway Plaza at UCLA and deploy it within a simulation environment. The workflow consists of image data collection of UCLA Gateway Plaza using a mobile phone while capturing the scene from a wide range of viewpoints. Then the images are processed through a reconstruction pipeline based on COLMAP [9] and 3D Gaussian Splatting [4] to generate a photorealistic representation of the environment. Finally, the reconstructed scene is post-processed and integrated into NVIDIA Isaac Sim [5], where it can be used for interactive visualization and simulation.

The resulting environment provides a realistic testbed for autonomous systems research. In particular, it enables the safe evaluation of autonomous vehicle perception, planning, and control algorithms within a virtual replica of a real-world urban intersection. The primary contributions of this project are: (1) the development of an end-to-end reconstruction pipeline for generating a digital twin of UCLA Gateway Plaza and (2) the integration of the reconstructed environment into NVIDIA Isaac Sim.

## 2 Methods

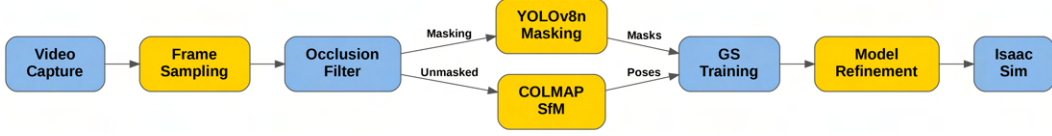


Figure 2: Full 3D Simulation Pipeline

Figure 2 illustrates our end-to-end 3D reconstruction and simulation pipeline. High-resolution footage (1080p at 30 frames per second) is first captured via smartphone and downsampled to 5 frames per second to reduce computational overhead and frame redundancy. To maintain photometric consistency, an automated occlusion filter evaluates the dataset and removes frames obstructed by large, monotone features, such as close-up foliage or hands. The filtered dataset then undergoes a dual-path preprocessing stage: YOLOv8n generates segmentation masks to exclude dynamic objects, while COLMAP operates strictly on the unmasked frames to guarantee stable feature matching. The resulting camera poses and masks are fed into NeRFStudio’s SplatFacto model [10] for 3D Gaussian Splatting optimization. Following reconstruction, the raw scene is pruned and reoriented using SuperSplat. Finally, the refined model is exported to NVIDIA Isaac Sim and layered over a hidden physical collision mesh to support realistic rigid-body dynamics.

### 2.1 Proof of Concept: Static Object Reconstruction

Before attempting the full plaza reconstruction, which involves a large, uncontrolled outdoor environment with dynamic pedestrians and vehicles, we validated the pipeline on a single parked vehicle using Google Colab’s cloud compute resources. This simple, isolated vehicle offered strong feature correspondences, allowing us to identify failure modes easily, especially when testing COLMAP and masking strategies.

**Masking Strategy and Optimization:** In unmasked reconstruction, the GS optimizer naturally attempts to reconstruct the entire scene alongside scattered floaters, which are noisy Gaussians produced by background clutter. We initially used the rembg [2] tool to replace the background with solid black pixels, but this caused the GS optimization to ignore background COLMAP initializations causing clutter. By switching to alpha-channel masks, we explicitly force the model to make all background gaussians transparent, which pruned them from the model. This eliminated background COLMAP-initialization floaters and confined the Gaussians entirely to the target vehicle.



(a) RemBG Background Mask Example



(b) Unmasked

(c) Naive Binary Masking

(d) Alpha-Channel Masking

Figure 3: Masking Experimentation on Proof-of-Concept

To determine the best capture modality, a single continuous video was initially captured by orbiting the vehicle. We subsequently tested a multi-video approach by recording the vehicle both inside a garage and outdoors to maximize coverage. However, differences in lighting and exposure, combined with the absence of shared viewpoints, led to cross-video misalignment and offset geometry. While

this does not completely rule out using multiple video viewpoints in the Gateway Plaza reconstruction, it established that a single continuous capture is more efficient and tractable for COLMAP than multi-video stitching.



Figure 4: Multi-Video Object Reconstruction Attempt

## 2.2 Scaling to UCLA Gateway Plaza: Data Capture Strategy

Due to time constraints and cloud compute limitations, the full plaza pipeline was migrated to a local GPU hosted on the UCLA Mobility Lab’s virtual machine. The entire pipeline took approximately 10 hours to complete.

Transitioning to the massive outdoor environment of UCLA Gateway Plaza required strict route planning, as opposed to the easy circumference route of the vehicle reconstruction. Due to the limitations of using a mobile phone camera rather than a drone, which lacks elevated perspectives and native high-precision GPS tracking, we relied entirely on visual overlap for feature co-location. We also found that third-party GPS tracking applications introduced massive annotation overhead that far exceeded the computational cost of purely visual feature matching.

After numerous trials, we established that filming on a Sunday afternoon provided optimal conditions. The overhead sun minimized shifting shadows, and the weekend timing drastically reduced dynamic occlusions from pedestrians and vehicles. To ensure COLMAP could map the massive scale, we maintained a continuous 5-minute and 21-second video capture.

Our optimized walking route involved moving slowly around the crosswalk perimeter, panning the camera to keep distinct details around the plaza’s center in frame, and holding the camera at different elevations to provide more parallax viewpoints for reconstruction. Crucially, walking directly through the center of the crosswalk was found to be necessary to capture adequate textures and viewing angles of the road surface, which otherwise suffered from poor reconstruction quality.

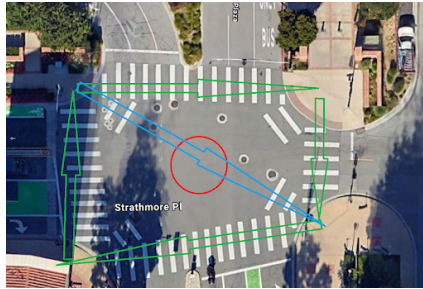


Figure 5: Data Capture Route

## 2.3 Preprocessing and Segmentation

The raw footage was sampled at 5 frames per second. This provided an optimal balance, supplying COLMAP with enough visual overlap for robust feature extraction while discarding redundant frames.

To handle the highly dynamic nature of the plaza, the simple `rembg` masking from our proof of concept was upgraded to the YOLOv8n segmentation model [3]. We specifically targeted and generated alpha-channel masks for pedestrians, moving vehicles, and buses. By hiding these dynamic occlusions, we

ensured the subsequent Structure-from-Motion (SfM) pipeline only registered static environmental features.

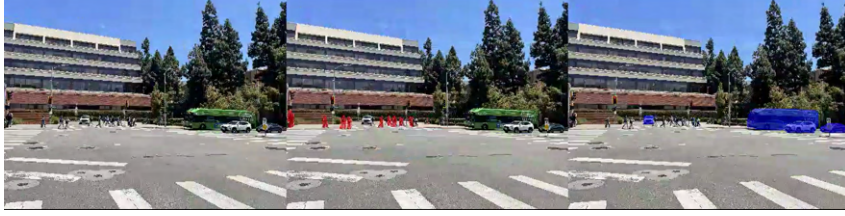


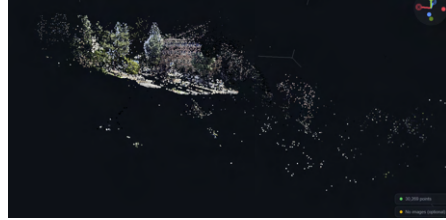
Figure 6: YOLOv8n Segmentation Masking

## 2.4 Structure-from-Motion and Point Cloud Initialization

To generate the sparse point cloud, camera intrinsics were automatically inferred from the mobile phone’s photo metadata and image size by COLMAP. We experimented with sequential matching, multi-view videos with spatial (GPS) matching, and vocabulary-tree matching to speed up the process. However, none of these methods could rival the capability and robustness of COLMAP’s *exhaustive matcher* for this specific dataset. Despite the longer runtime, exhaustive matching produced a highly comprehensive and successful point-cloud initialization, completely avoiding the severe warping, missing viewpoints, and large geometric gaps that characterized failed matching attempts.



(a) Successful Exhaustive Matching



(b) Intermediate Failed COLMAP Point Cloud

## 2.5 3DGS Training and Model Refinement

The unmasked images, sparse COLMAP point cloud, and YOLOv8n alpha masks were fed into the Splatfacto model within the NeRFStudio framework. Training the model for 60,000 iterations was sufficient to converge on a highly photorealistic scene. The raw model was then exported as a .ply file and imported into SuperSplat [8]. In this viewer, we manually aligned the scene with global world coordinates and performed pruning steps as needed, deleting irrelevant, external Gaussians representing small distant features and sky.



(a) Splatfacto Training Viewer



(b) Product of Supersplat Editing

## 2.6 Simulation Setup and Isaac Sim Integration

The simulation environment was configured across two operating systems. The 3DGRUT [6] project framework was implemented on Ubuntu 20.04 to convert the high-fidelity Gaussian Splatting .ply



models into native Universal Scene Description (.usdc) format. The actual simulation was run using NVIDIA Isaac Sim on Windows 11, though Linux is recommended for optimal performance.

Because Isaac Sim does not natively support physical interactions with standard Gaussian Splatting point clouds, we utilized a dual-layer architecture. Geospatial data for the UCLA Gateway Plaza was fetched from the OpenStreetMap [7] dataset and converted into a structured 3D collision mesh using a dedicated Blender [1] plugin. As shown in Figure 9, the 3D mesh has perfect physical accuracy and visual appearance. Both the photorealistic .usdc assets and the structural OSM collision mesh were imported into Isaac Sim. Meticulous manual alignment was then performed to perfectly overlap the invisible 3D collision mesh with the visual landmarks of the 3DGS overlay.

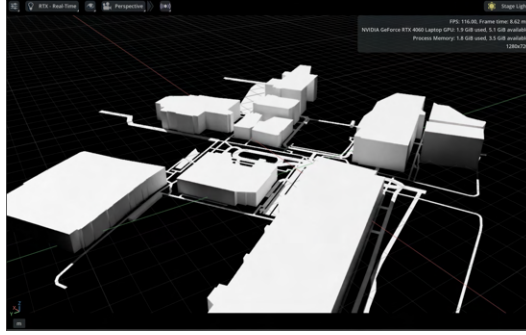


Figure 9: OpenStreetMap 3D Plaza Mesh

## 2.7 Vehicle Physics and Kinematic Control

To simulate autonomous traffic behavior, a 3D vehicle model was introduced. Physical properties were configured to facilitate realistic interactions by encapsulating the vehicle within a simplified rigid-body bounding box proxy, optimizing performance. Collider features were applied to the vehicle bounding box, the 3D structural mesh, and the ground plane, with gravity added to ensure ground contact.

For trajectory execution, we bypassed default PhysX ground mesh friction anomalies by implementing custom kinematic controllers using the Isaac Sim XFormPrim API. The controller ran at a 60Hz update rate to ensure precise, drift-free maneuvers. During straight driving, the Z-axis velocity was strictly locked to prevent suspension bouncing caused by minor mesh irregularities. For turning maneuvers, such as navigating a 90-degree left turn, pure geometric angular velocities were applied directly via quaternion-to-Euler transformations. Furthermore, rotational offset matrices were applied to the vehicle’s forward vector to compensate for the non-orthogonal alignment of the real-world plaza intersection relative to the global coordinate system, resulting in flawless trajectory execution.

## 3 Results

### 3.1 3D Reconstruction

Table 1: Final Reconstruction Pipeline Parameters for UCLA Gateway Plaza

| Parameter               | Value                 |
|-------------------------|-----------------------|
| Capture Duration        | 5 min 21 sec          |
| Capture Time            | Sunday, 1:00 PM       |
| Frame Rate              | 5 FPS                 |
| Total Frames            | 1605                  |
| COLMAP Matching Type    | Exhaustive            |
| 3DGS Iterations         | 60,000                |
| YOLOv8n Masking Targets | Pedestrians, Vehicles |

Due to the inherent 3D nature of our results, we can only show limited results through static images, so we have instead provided an interactive demonstration website and supersplat viewer for reference:

<https://uclagatewayplaza3dgs-f8wajulkkjqwy7nf4cusxg.streamlit.app/>

<https://superspl.at/scene/ccb8ad42>



Figure 10: Example 3DGS Rendered Frames of UCLA Gateway Plaza

As can be seen in both the website and Supersplat demos, alpha-masking frames to remove pedestrian and vehicular dynamic occlusions was incredibly effective and results in a perfectly clear reconstruction of the scene as if those objects never existed in the first place. This is largely accomplished due to the significant overlap achieved by our continuous video sampled at 5fps, allowing the full scene to be reconstructed despite momentary occlusions

We also report a full video flythrough, which spans many unseen viewpoints and emphasizes the structural and perceptual consistency of our reconstruction. We do note that road surfaces on side-streets adjacent to the crosswalk suffer from low density reconstruction, although this is expected, as our route intentionally focuses on the central crosswalk and surrounding buildings and traffic infrastructure.

### 3.2 NVIDIA Isaac Sim

Overall, the developed hybrid simulation pipeline successfully merged hyper-realistic 3DGS environments with rigid-body collision modeling, validated by the precise controlling of the test vehicle.

#### 3.2.1 Hybrid Model Construction

After our adjustments, the 3D physics mesh and the imported 3DGS have a high degree of overlap, which proves that our work on 3D Gaussian splatting is reliable.

As shown in Figure 11, the main road, sidewalk and buildings in the corner from 3DGS highly overlap the 3D structural mesh.

After we imported our vehicle into the hybrid model, the vehicle landed on the path and no visually through-modeling scene or ground clipping was observed.

#### 3.2.2 Kinematic Control

The kinematic control was executed using the custom XFormPrim controller, resulting in flawlessly smooth trajectories for both driving tasks. Because the standard simulation approaches would have suffered from orientation drift, and our imported model originally have offset angles with the World's coordinate system, our script must compensated for the inherent non-orthogonal map alignment

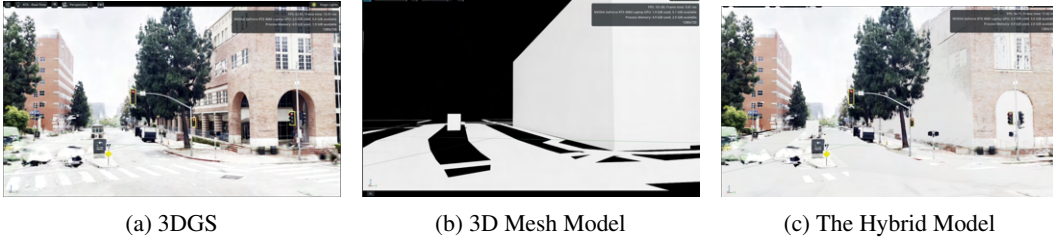


Figure 11: Overview of the Reconstruction of Hybrid Model in Isaac simulation.

through precise offset angle corrections to achieve the precise 90-degree turn, enabling a perfect centered path through the Westwood Plaza intersection.

We can clearly see that the vehicle is going straight, makes a 90-degree left turn in the center of the crossing, and drives away. To make it more clear, We remove the Visibility of 3DGS. This is shown in Figure 12.



Figure 12: The Success Kinematic Control in Simulation with XFormPrim Applied

## 4 Conclusion and Future Work

In our project, we developed a pipeline for creating a high-fidelity digital twin of UCLA Gateway Plaza and deployed it within NVIDIA Isaac Sim. Using a short mobile-phone video, we reconstructed a photorealistic 3D representation of the environment with COLMAP and 3D Gaussian Splatting, while applying YOLO-based masking to remove pedestrians and vehicles. The resulting reconstruction exhibits geometric consistency and realistic rendering from captured and uncaptured viewpoints. We then integrated the reconstructed scene into NVIDIA Isaac Sin to create a hybrid simulation environment that combines the realistic scene with physics-based interactions through collision meshes.

The pipeline developed integrates modern neural reconstruction methodology with robotics simulation tools to create a realistic digital twin. It serves as a valuable testbed for autonomous vehicle testing while reducing the cost, risk, and challenges associated with real-world testing.

Due to limited computational resources during the initial stages of this project, we focused on a traditional Gaussian Splatting workflow. Future work could investigate feed-forward reconstruction methods such as DepthSplat, which can significantly reduce training time while maintaining high-quality scene representations. Additionally, distributed 3D Gaussian Splatting techniques allows for the use of multiple GPUs, while diffusion-based enhancement methods such as Difix3D+ may further improve rendering quality and visual consistency. From a data collection perspective, we could incorporate multiple videos recorded from different viewpoints. Increased coverage may improve reconstruction scale and completeness. Finally for the simulation, we could introduce multiple vehicles to simulate complex, real-world traffic intersections as well as optimize the vehicle's turning and braking deceleration procession to more closely resemble real-world scenarios.

## A GitHub Demonstration Platform

[https://github.com/koroltony/UCLA\\_Gateway\\_Plaza\\_3DGS](https://github.com/koroltony/UCLA_Gateway_Plaza_3DGS)

## B Website Link and Information

<https://uclagatewayplaza3dgs-f8wajulkjqwy7nf4cusxg.streamlit.app/>

## C SuperSplat Interactive Page

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